

EDA Process Overview Questions

Conceptual Questions

- 1.What is the primary goal of Exploratory Data Analysis (EDA) and why is it considered a crucial step before building machine learning models?**
- 2.List and explain the 5 main phases of the EDA process. What specific activities are performed in each phase?**
- 3.Compare and contrast the roles of descriptive statistics and data visualization in EDA. When would you prefer one over the other?**
- 4.What are the key differences between univariate, bivariate, and multivariate analysis? Provide examples of when you would use each.**
- 5.Explain how EDA helps in identifying data quality issues. What are the common data quality problems you typically look for?**

Practical Application Questions

- 6.Given a new dataset, describe your step-by-step approach for the initial data inspection. What specific Python functions would you use and what information would you look for?**
- 7.You discover that 30% of the values in a critical numerical column are missing. What strategies would you employ to investigate the pattern of these missing values and decide on the appropriate handling method?**
- 8.How would you create a comprehensive EDA report template that can be reused across different projects? What sections would you include and why?**
- 9.Describe three different methods to identify outliers in a dataset and explain the pros and cons of each method.**
- 10.You have both numerical and categorical variables in your dataset. How would your EDA approach differ for each data type?**

Practical Application Questions

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Scenario-Based Questions

11. **You're analyzing customer data for an e-commerce company. The dataset has 50,000 rows and 25 columns including customer demographics, purchase history, and website behavior. Outline your complete EDA plan.**
12. **During EDA, you find that two variables show a correlation coefficient of 0.85. What additional analyses would you perform to understand this relationship better and avoid spurious conclusions?**
13. **Your dataset has datetime columns spanning 3 years. How would you incorporate time-series analysis into your EDA process?**
14. **You're working with a dataset that has high-dimensional data (100+ features). What specialized EDA techniques would you use to handle this complexity?**
15. **After completing EDA, stakeholders ask you to justify the time spent on this phase. How would you demonstrate the value and ROI of your EDA efforts?**